

Amulet Motion Controller

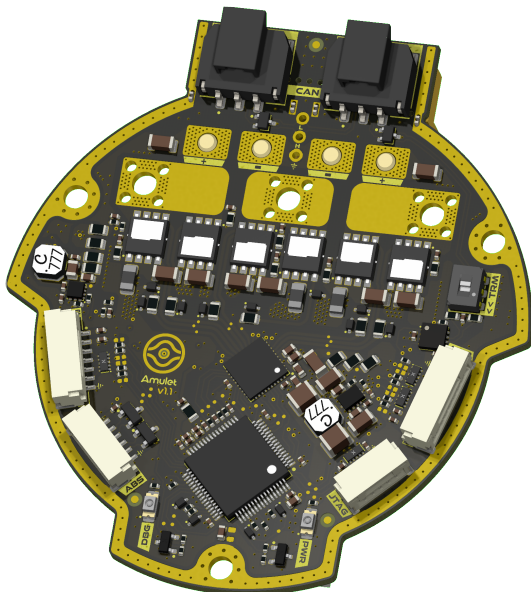
Variant: RELEASED

2024-11-20

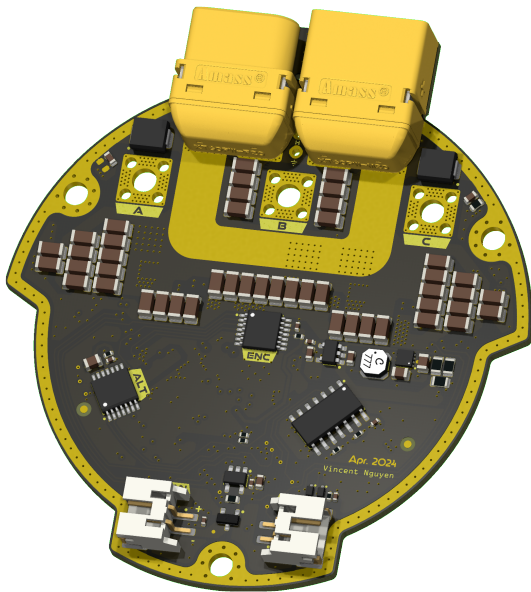
Rev 1.1

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TOP VIEW



BOTTOM VIEW



DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

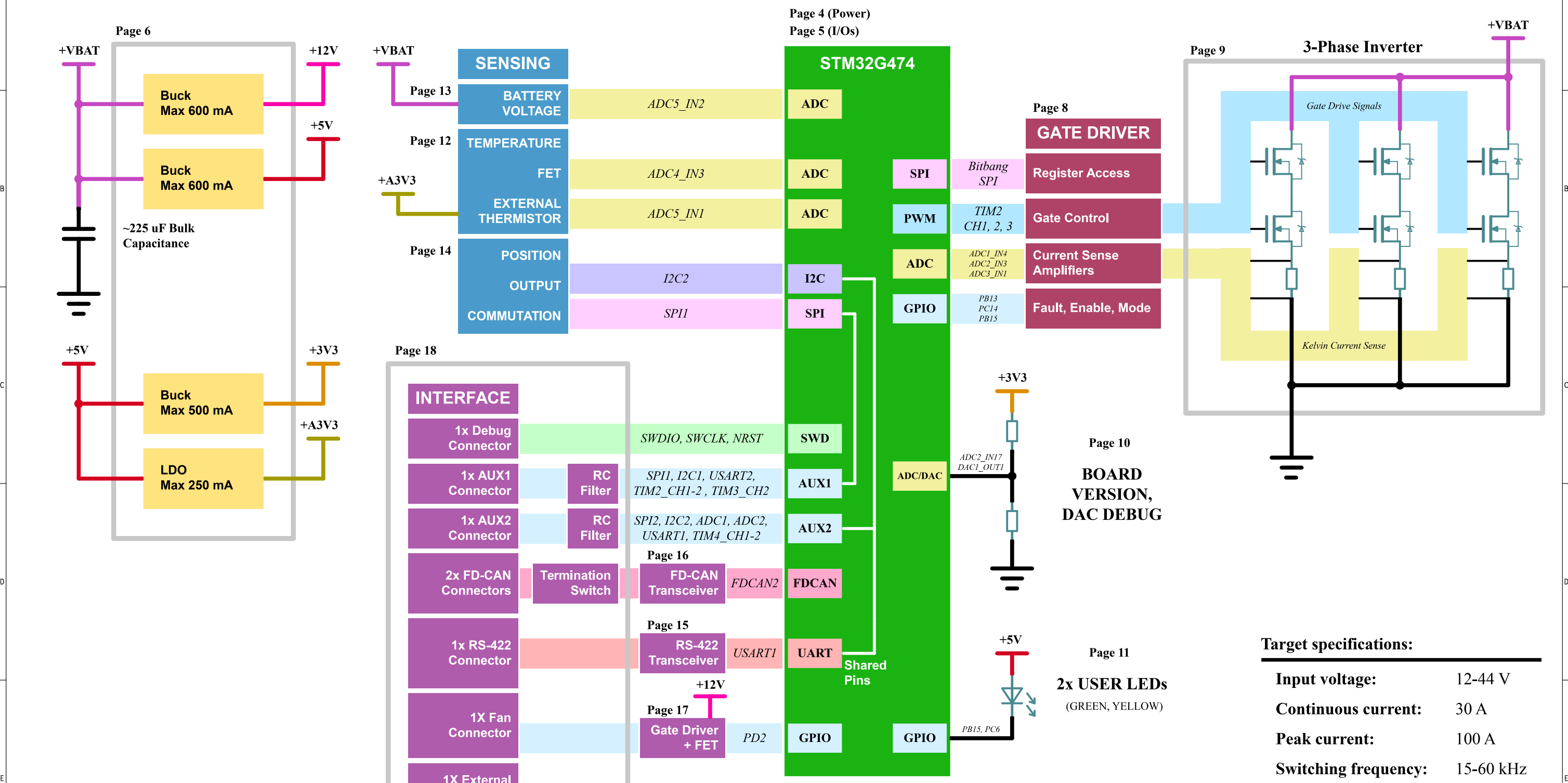
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 13-APR-2024

	Comments:	Company: EPFL Xplore Research			Variant: RELEASED	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller			Project Name: Chienpanzé	
	Sheet Title: Cover Page	File Name: amulet_controller.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1	
	Sheet Path: /		Reviewer:		Size: A3	Sheet: 1 of 21

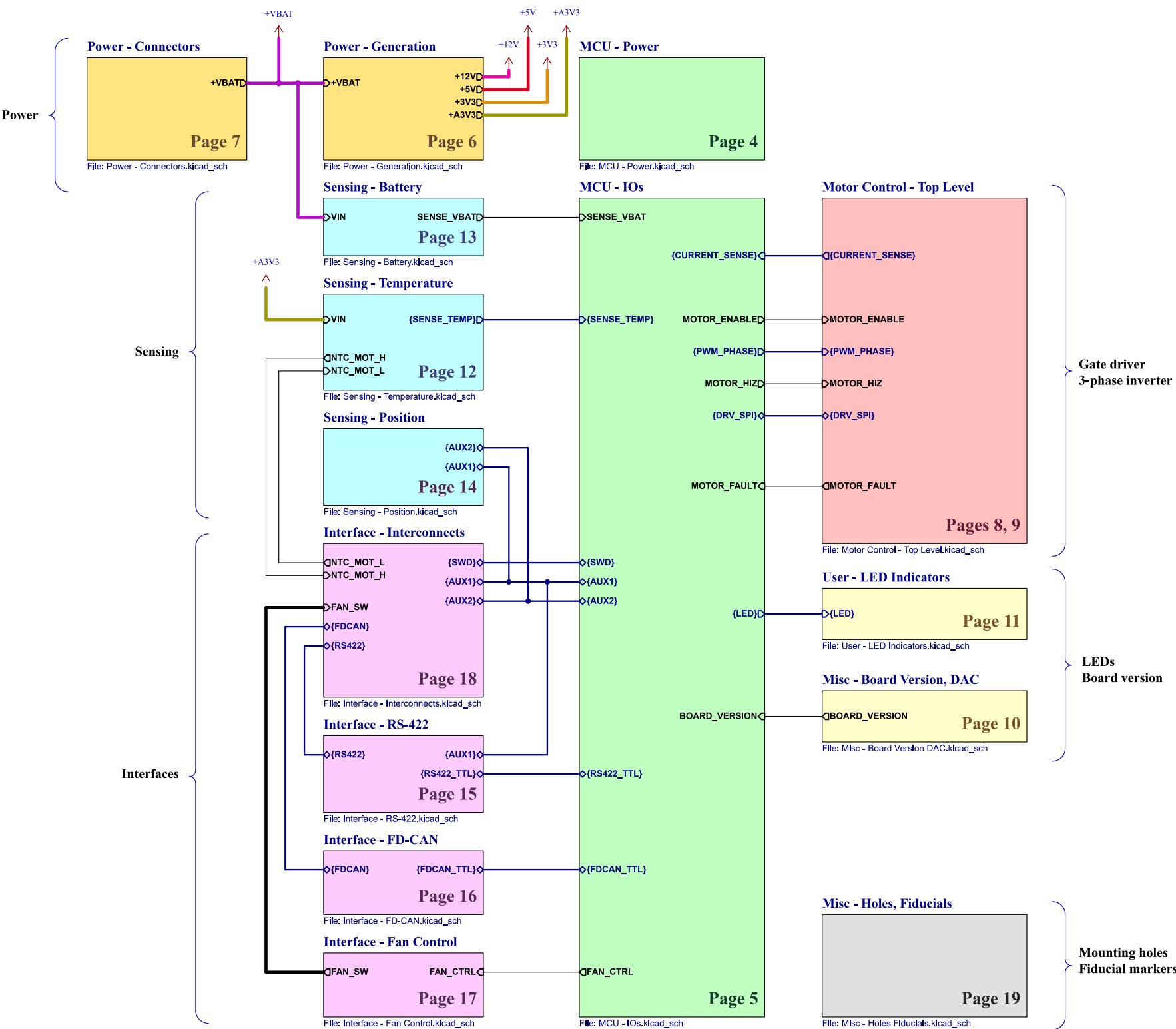
[2] Block Diagram



Target specifications:	
Input voltage:	12-44 V
Continuous current:	30 A
Peak current:	100 A
Switching frequency:	15-60 kHz

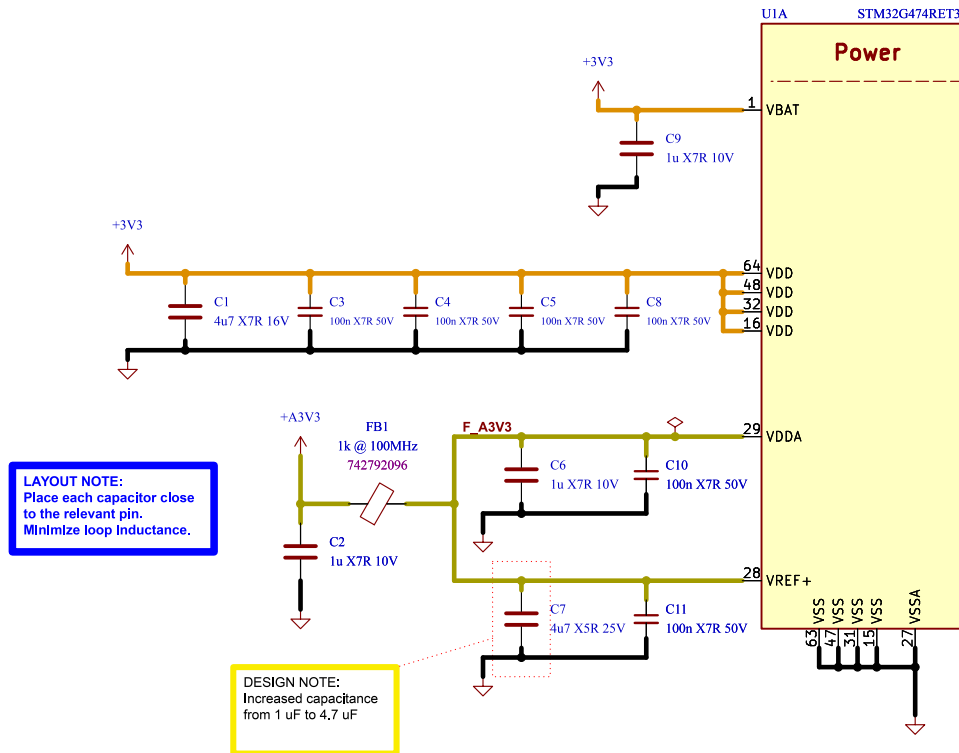
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		Board Name: Amulet Motion Controller			Project Name: Chienpanzé	
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1	
	Sheet Path: /Block Diagram/		Reviewer:		Size: A3	Sheet: 2 of 21

[3] Project Architecture



Comments:	Company: EPFL Xplore Research			Variant: RELEASED	Git Hash: 04d3b06
	Board Name: Amulet Motion Controller			Project Name: Chienpanzé	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-22	Revision: 1.1	
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 21	

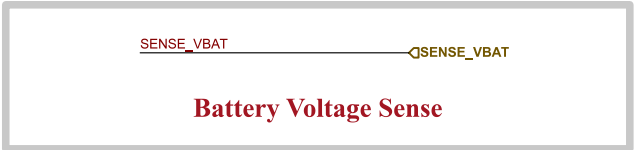
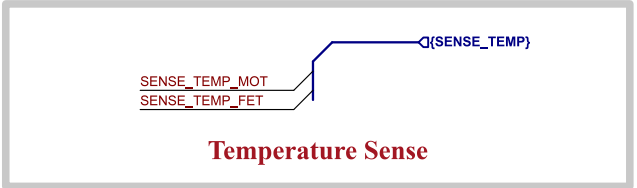
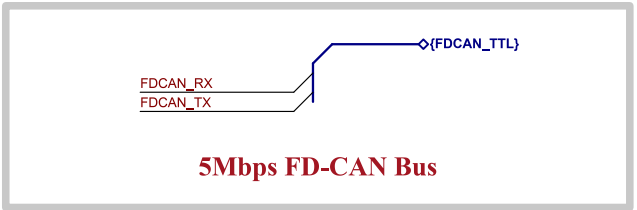
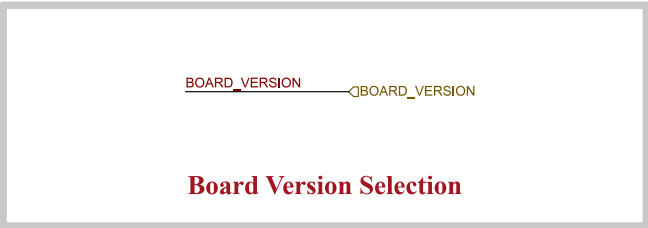
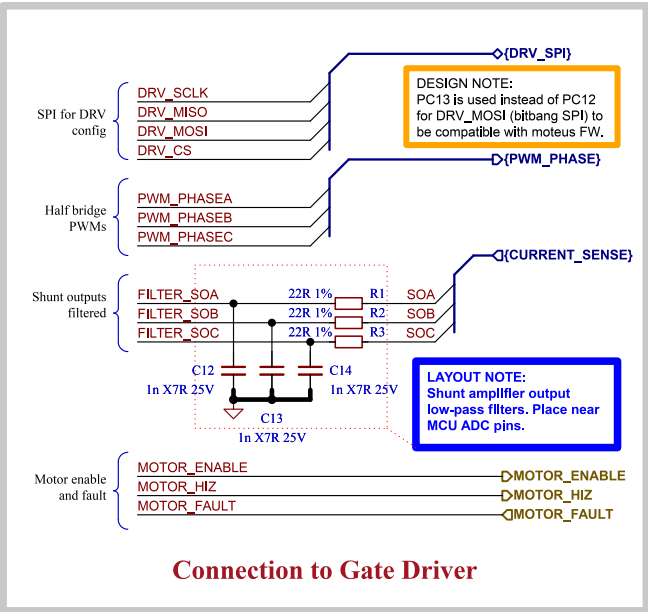
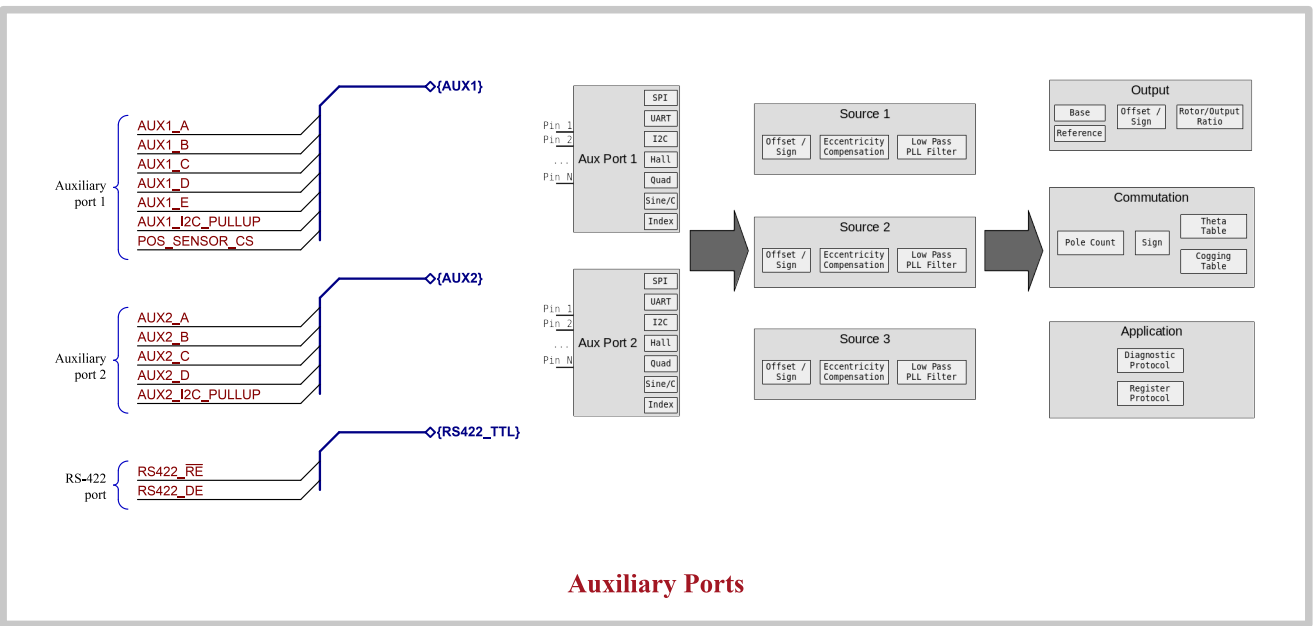
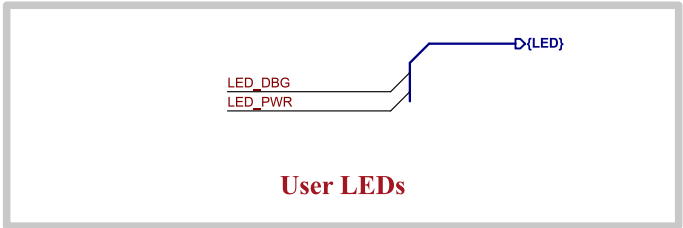
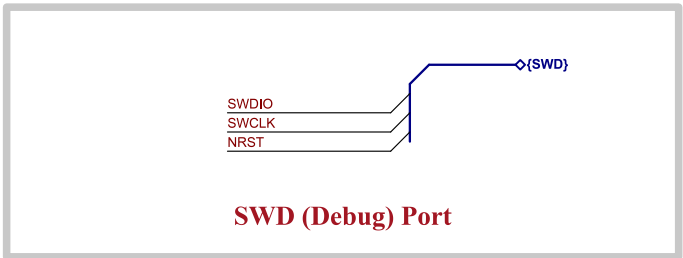
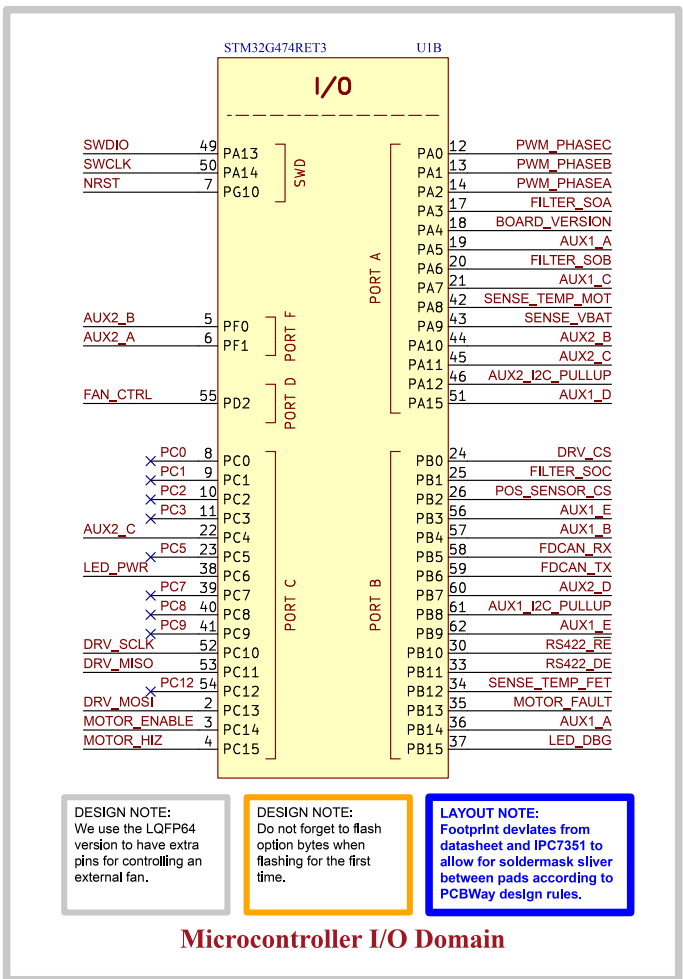
[4] MCU - Power



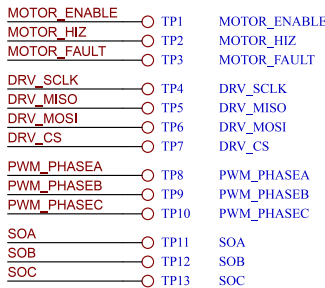
Microcontroller Power Supply Scheme

	Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: EPFL Xplore Research		Variant: RELEASED	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: MCU - Power	File Name: MCU - Power.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-18	Revision: 1.1
	Sheet Path: /Project Architecture/MCU - Power/		Reviewer:	Size: A4	Sheet: 4 of 21

[5] MCU - I/Os



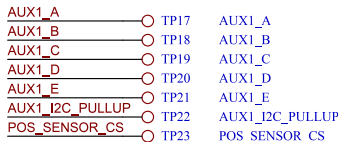
Gate Driver



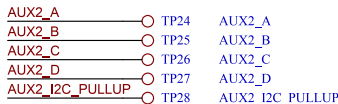
Debug



Auxiliary pins 1



Auxiliary pins 2



RS-422



LEDs



FD-CAN



Fan



Sense



Test Points

Comments:

References:

Flexible I/O worked examples
Flexible I/O source configuration

Sheet Title:

MCU - I/Os

Sheet Path:

/Project Architecture/MCU - I/Os/

Company:

EPFL Xplore Research

Board Name:

Amulet Motion Controller

File Name:

MCU - I/Os.kicad_sch

Designer:

Vincent Nguyen

Reviewer:



Variant:

RELEASED

Project Name:

Chienpanzé

Date:

2023-12-20

Size:

A3

Git Hash:

04d3b06

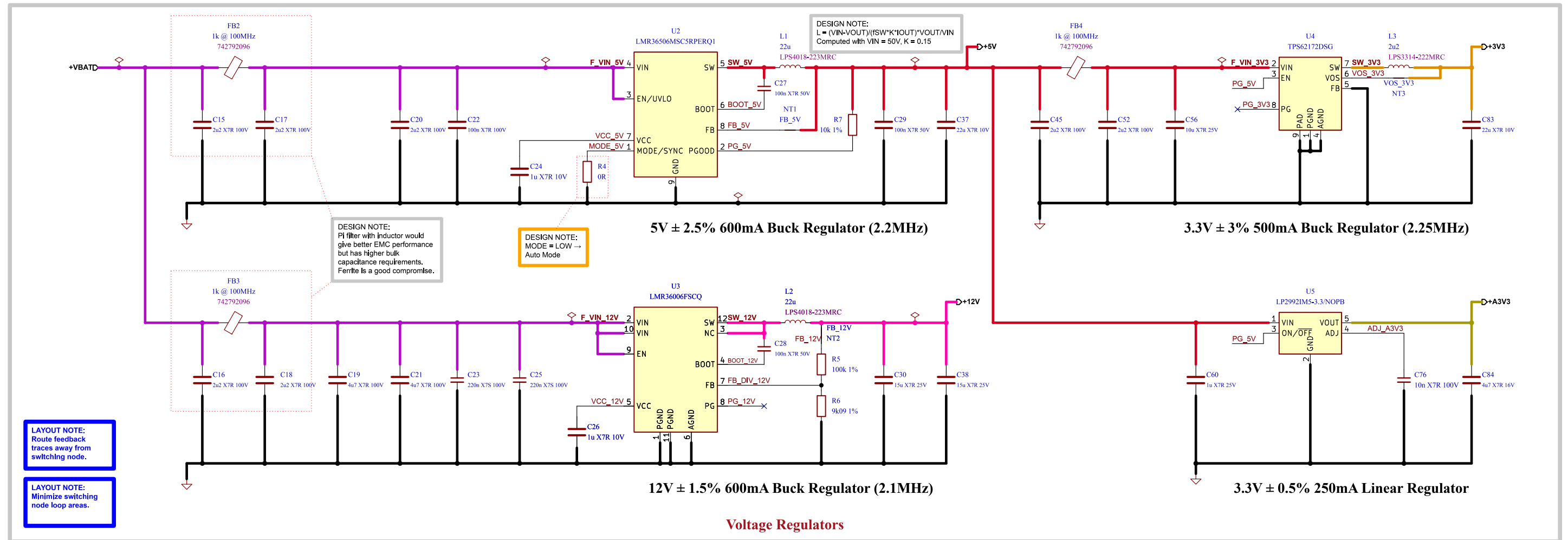
Revision:

1.1

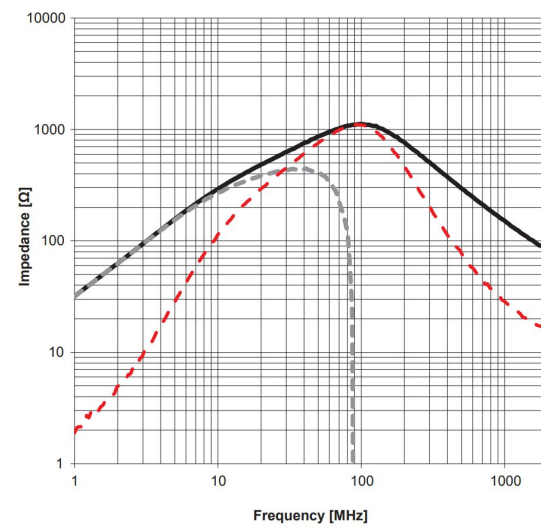
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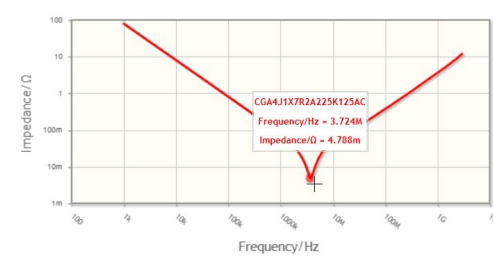
[6] Power - Generation



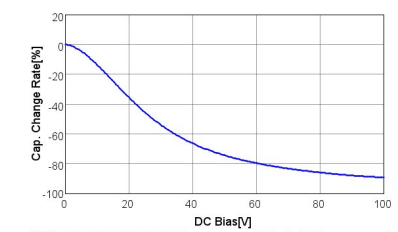
Ferrite bead Impedance vs Frequency curve



Pi filter capacitor Impedance vs Frequency curve



Bulk capacitance Derating curve



Capacitor and Ferrite Bead Parameters

Battery

+VBAT TP39 +VBAT
+VBAT TP40 +VBAT
+VBAT TP41 +VBAT

Filtered Power Rails

F_VIN_12V TP42 F_VIN_12V
F_VIN_5V TP43 F_VIN_5V
F_VIN_3V3 TP44 F_VIN_3V3

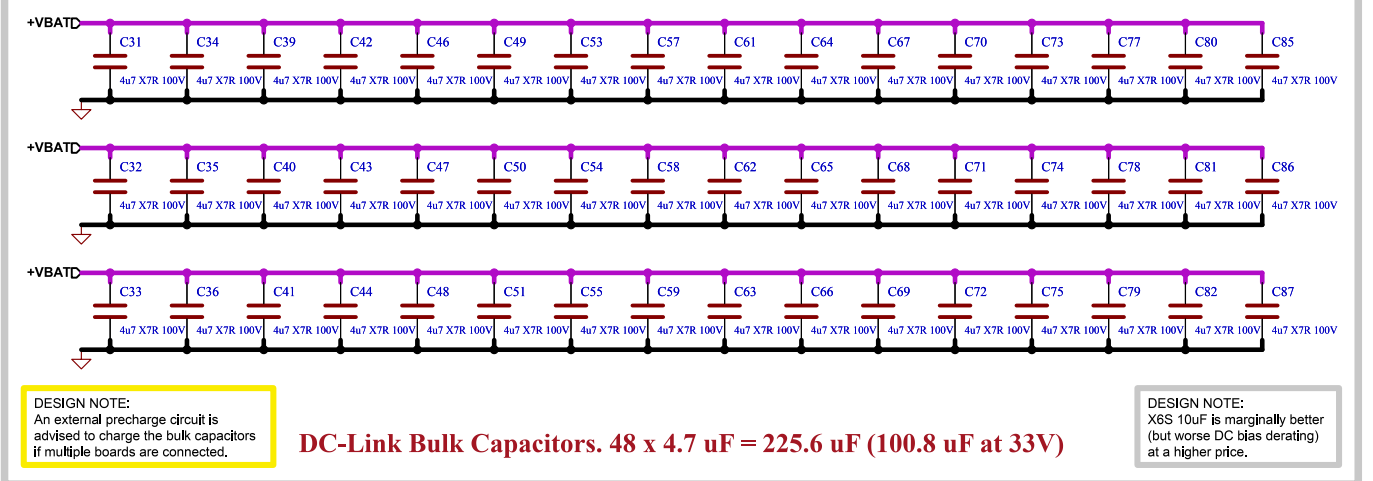
Power Rails

+12V TP45 +12V
+5V TP46 +5V
+5V TP47 +5V
+3V3 TP48 +3V3
+3V3 TP49 +3V3
+3V3 TP50 +3V3
+A3V3 TP51 +A3V3

Ground

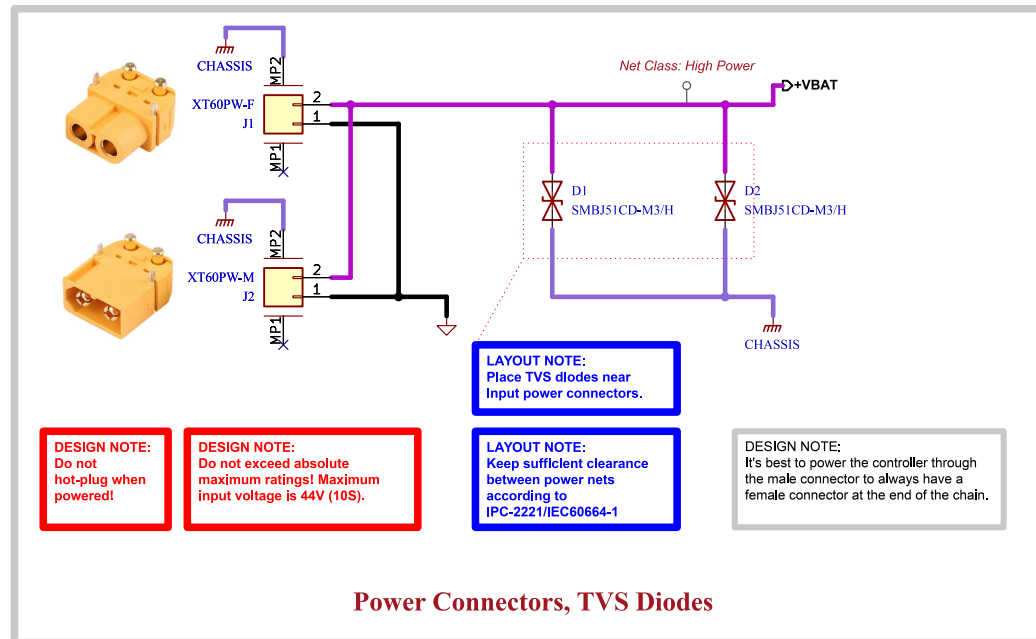
GND TP52 GND
GND TP53 GND
GND TP54 GND
GND TP55 GND
GND TP56 GND
GND TP57 GND
GND TP58 GND

Test Points



Comments: +12V (600 mA): Fan power +5V (600 mA): Gate Driver, LEDs +3V3 (500 mA): digital 3.3V +3V3A (250 mA): analog 3.3V	Company: EPFL Xplore Research		Variant: RELEASED	Git Hash: 04d3b06
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Power - Generation	File Name: Power - Generation.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
Sheet Path: /Project Architecture/Power - Generation/	Reviewer:		Size: A3	Sheet: 6 of 21

[7] Power - Connectors



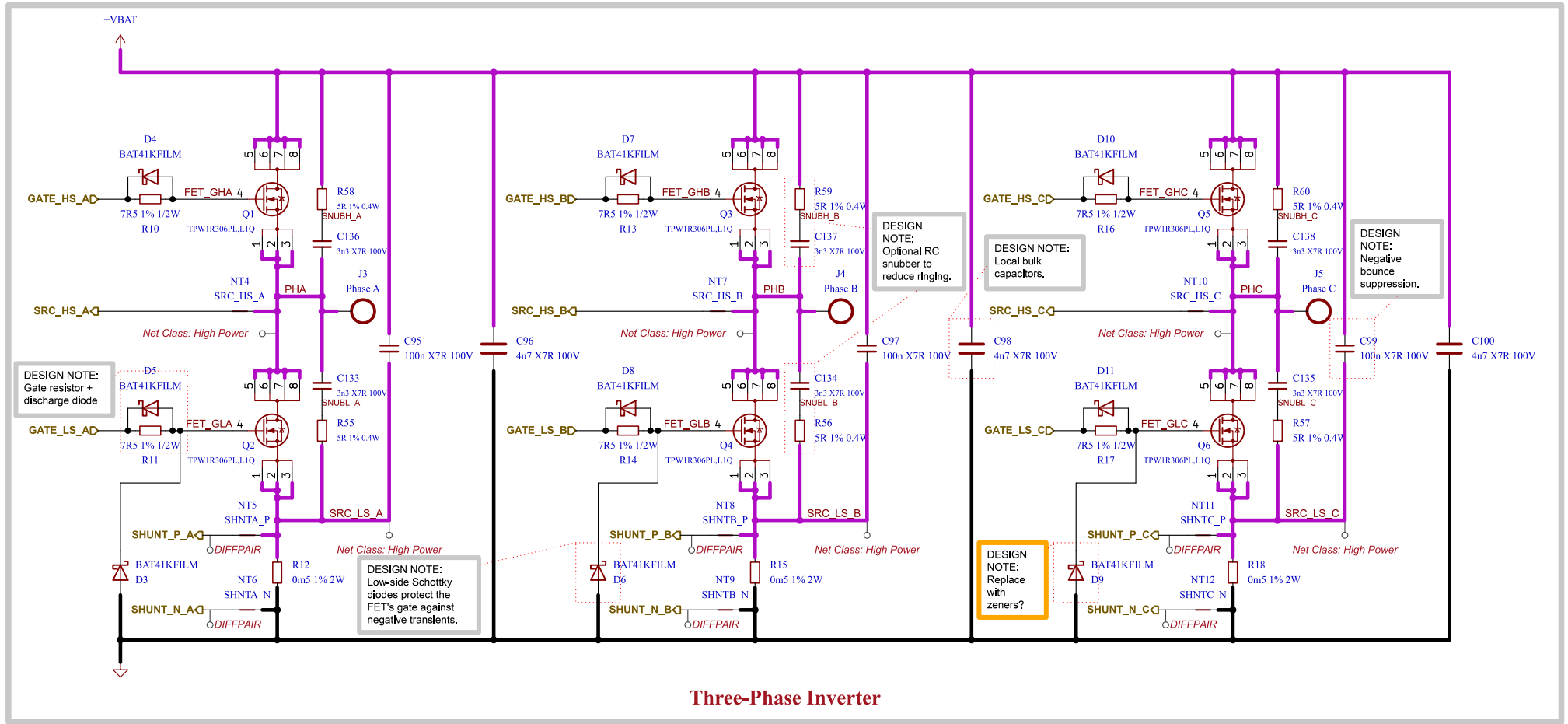
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		Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
	Sheet Title: Power - Connectors	File Name: Power - Connectors.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31
	Sheet Path: /Project Architecture/Power - Connectors/	Reviewer:	Size: A4	Revision: 1.1
			Sheet: 7	of 21

100-V Three-Phase Smart Gate Driver



	Comments:	Company:			Variant:	RELEASED	Git Hash:	04d3b06
		Board Name:	Amulet Motion Controller			Project Name: Chienpanzé		
	Sheet Title:	File Name:	Designer:	Date:	Revision:			
	Motor Control - Top Level	Motor Control - Top Level.kicad_sch	Vincent Nguyen	2023-12-20	1.1			
	Sheet Path:			Reviewer:	Size:	Sheet:		
	/Project Architecture/Motor Control - Top Level/				A3	8 of 21		

[9] Motor Control - Inverter



LAYOUT NOTE:
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

DESIGN NOTE:
A gate drive current that is too large can damage the FETs!

Comments:
System Design Considerations for High-Power Motor Driver Applications
Best Practices for Board Layout of Motor Drivers
Proper RC Snubber Design for Motor Drivers

Sheet Title:
Motor Control - Inverter

Sheet Path:
/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

Company:
EPFL Xplore Research



Board Name:
Amulet Motion Controller

File Name:
Motor Control - Inverter.kicad_sch

Reviewer:

Designer:
Vincent Nguyen

Variant:
RELEASED

Project Name:
Chienpanzé

Date:
2024-01-25

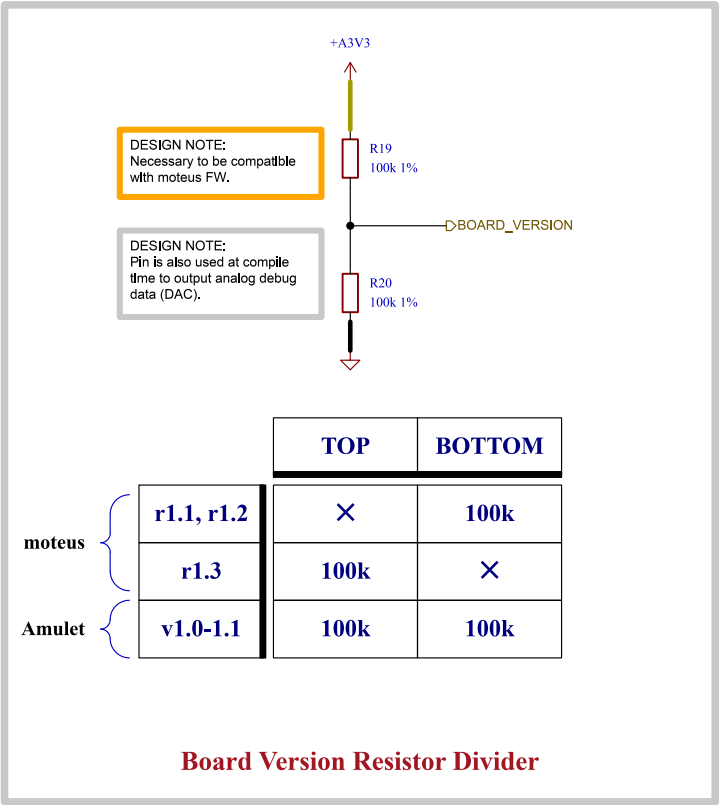
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Git Hash:
04d3b06

Revision:
1.1

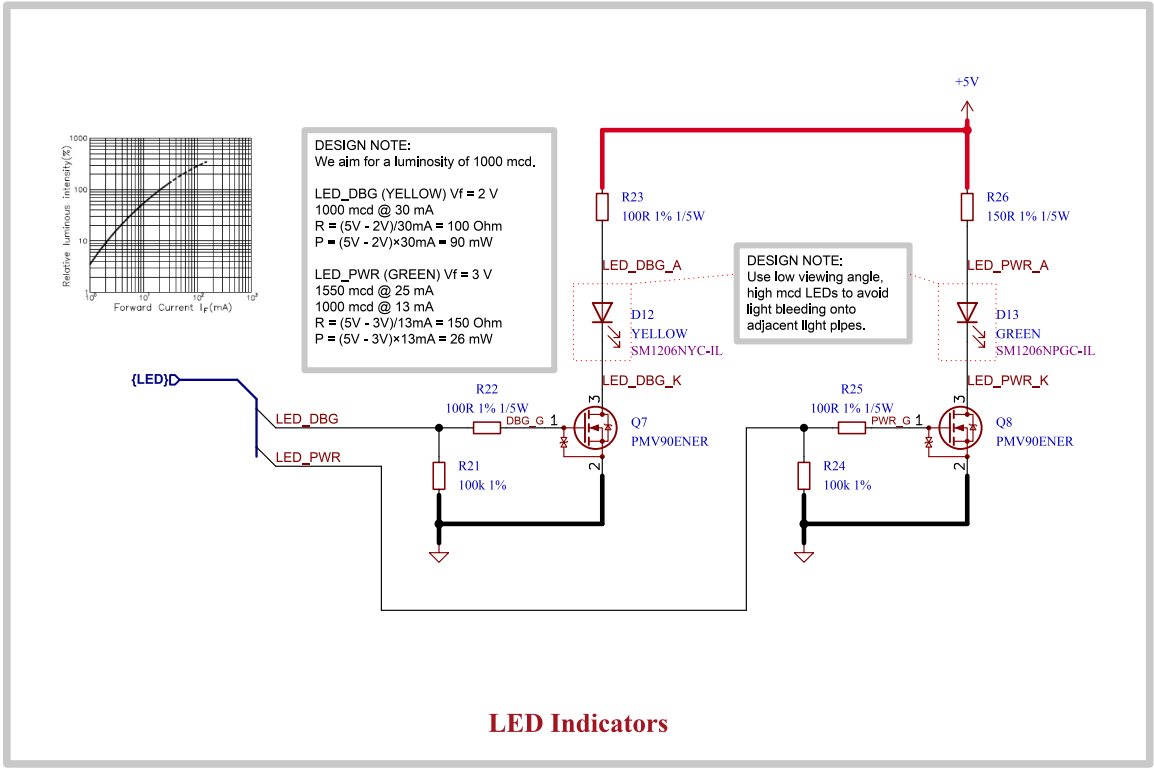
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[10] Misc - Board Version, DAC



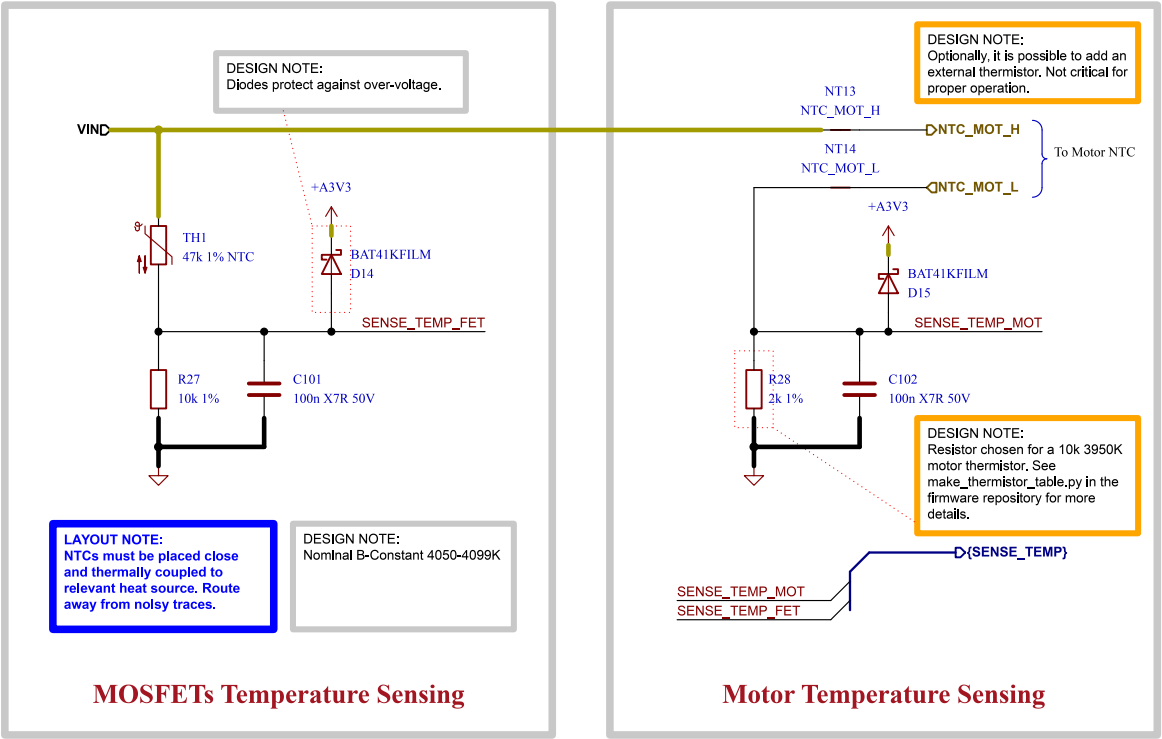
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	Sheet Title: Misc - Board Version, DAC	File Name: Misc - Board Version DAC.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
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[11] User - LED Indicators



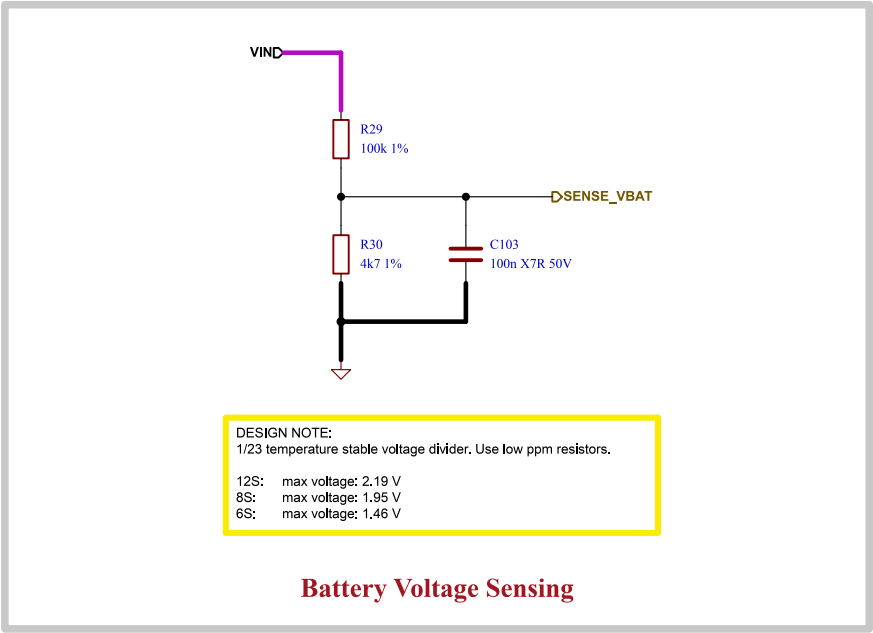
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	Sheet Path: /Project Architecture/User - LED Indicators/		Reviewer:	Size: A4	Sheet: 11 of 21

[12] Sensing - Temperature



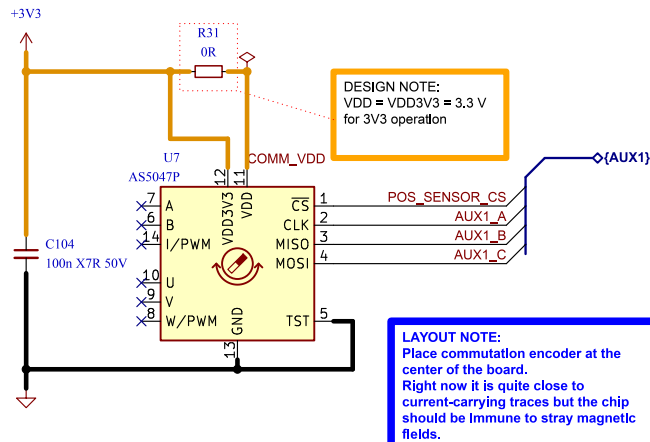
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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Sensing - Temperature	File Name: Sensing - Temperature.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
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[13] Sensing - Battery



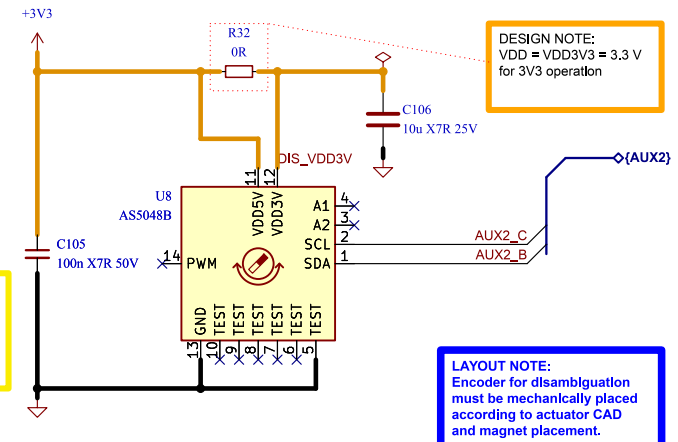
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	Sheet Title: Sensing - Battery	File Name: Sensing - Battery.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-14	Revision: 1.1
	Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer:	Size: A4	Sheet: 13 of 21

[14] Sensing - Position



14-Bit On-Axis Magnetic Rotary Position Sensor (commutation)

DESIGN NOTE:
AS5047P senses magnet mounted on planetary sun gear, for commutation.
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.



14-Bit On-Axis Magnetic Rotary Position Sensor (disambiguation)

DESIGN NOTE:
The encoder for disambiguation cannot be used to estimate backlash!

LAYOUT NOTE:
Encoder for disambiguation must be mechanically placed according to actuator CAD and magnet placement.

Comments:

Company:

EPFL Xplore Research



Variant:

RELEASED

Git Hash:

04d3b06

Board Name:

Amulet Motion Controller

Project Name:

Chienpanzé

Sheet Title:

Sensing - Position

File Name:

Sensing - Position.kicad_sch

Designer:

Vincent Nguyen

Date:

2023-10-14

Revision:

1.1

Sheet Path:

/Project Architecture/Sensing - Position/

Reviewer:

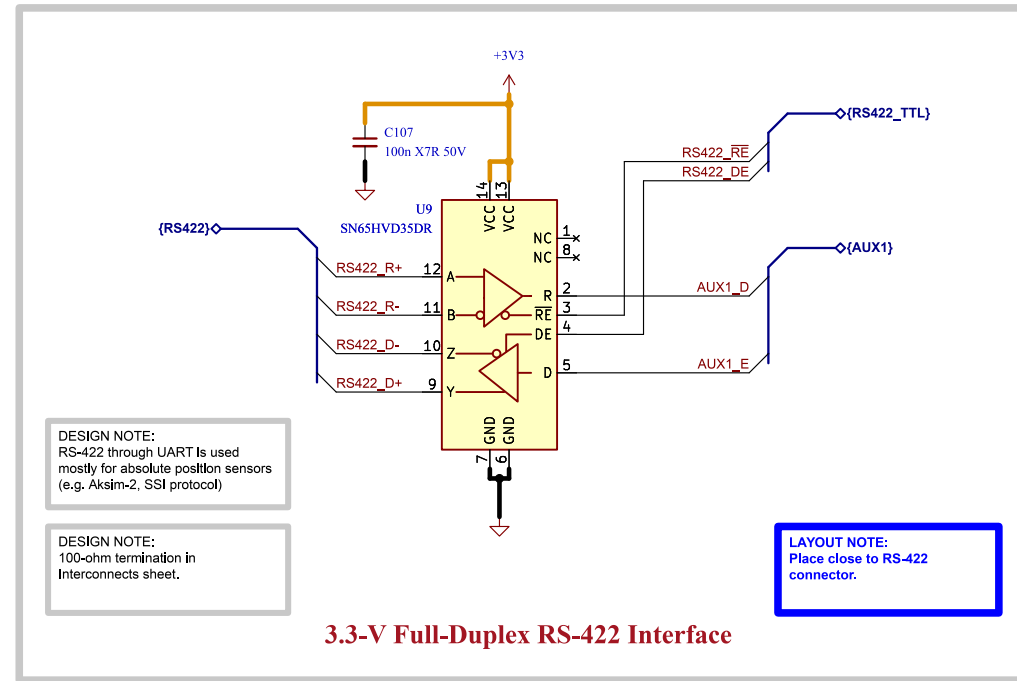
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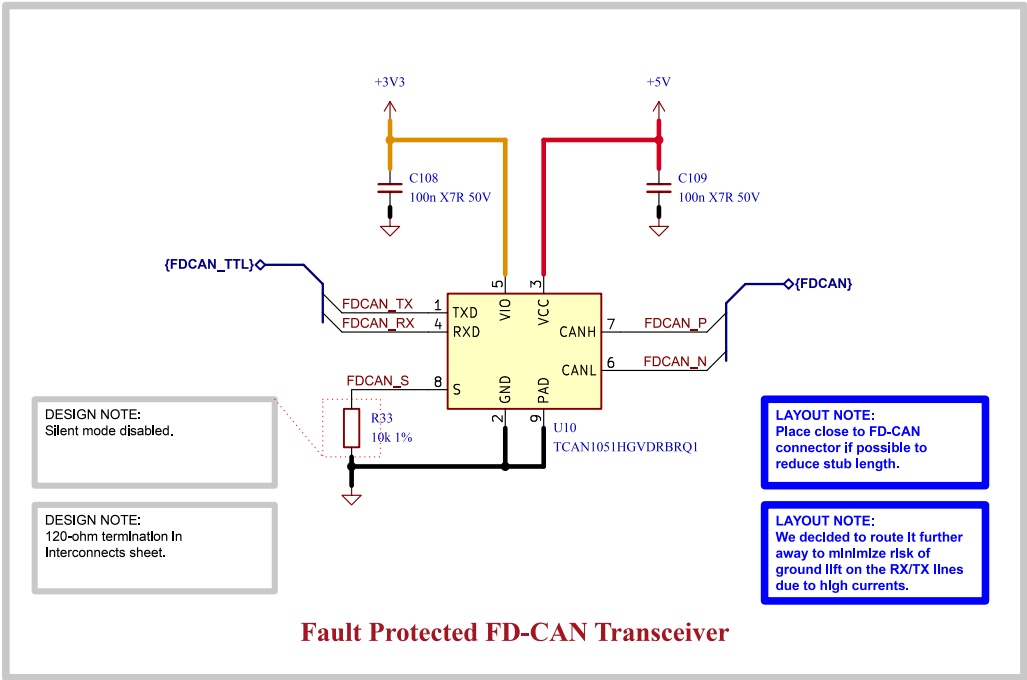
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[15] Interface - RS-422



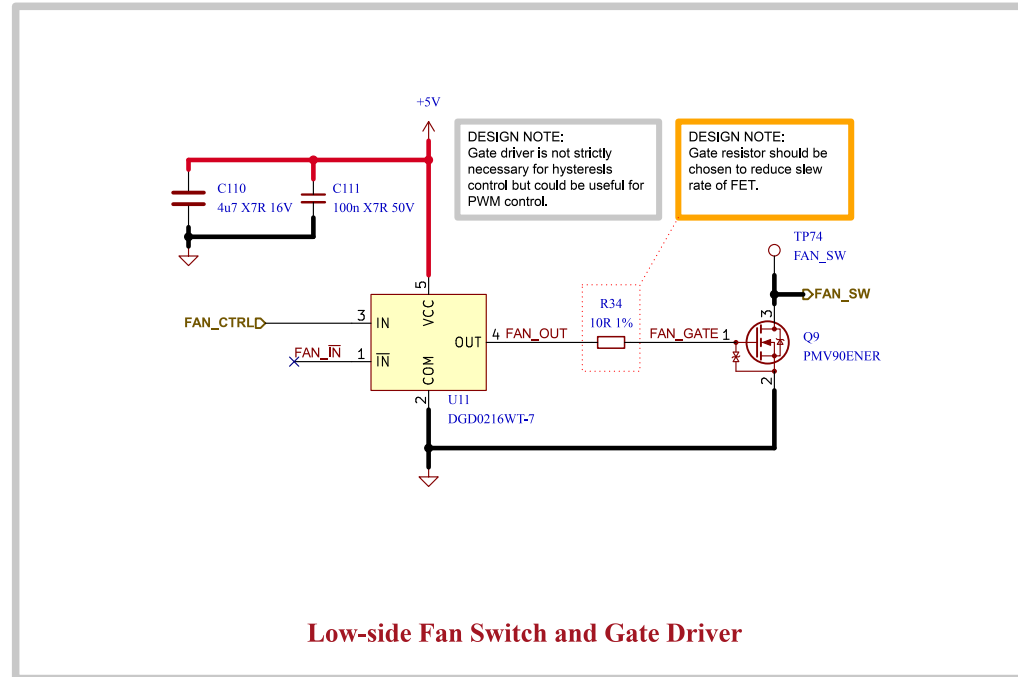
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[16] Interface - FD-CAN



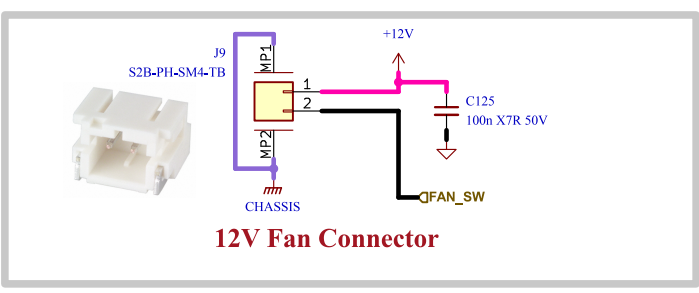
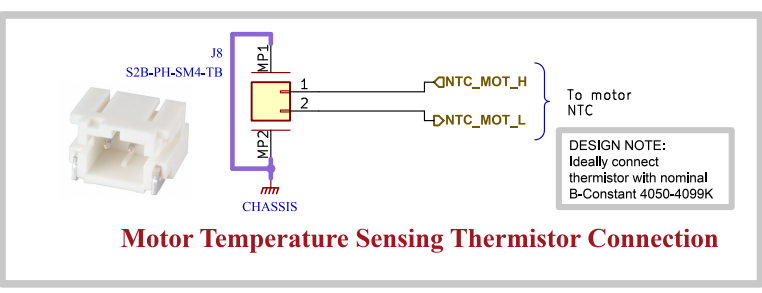
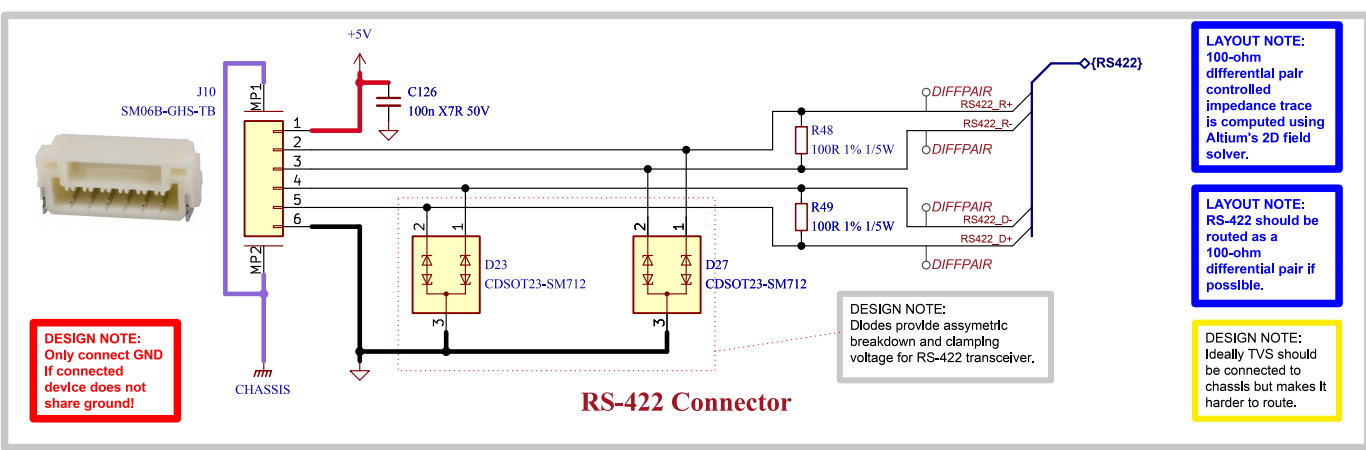
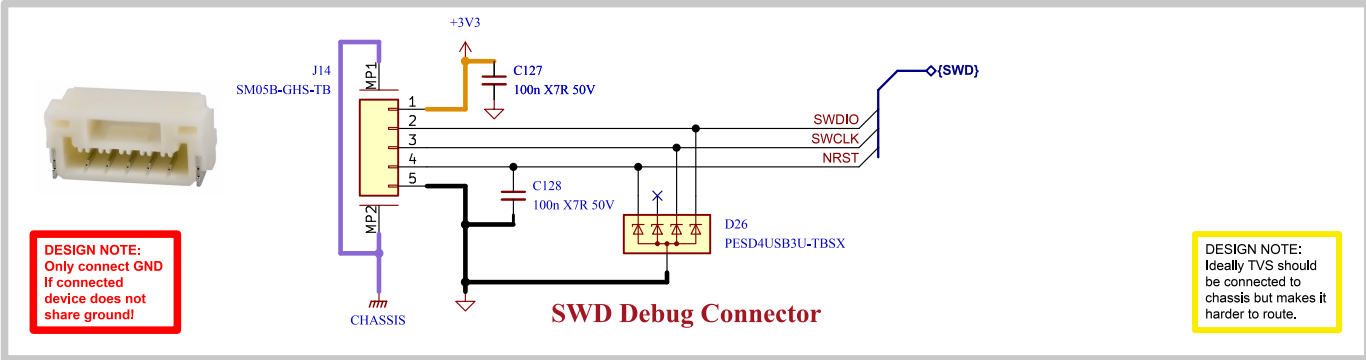
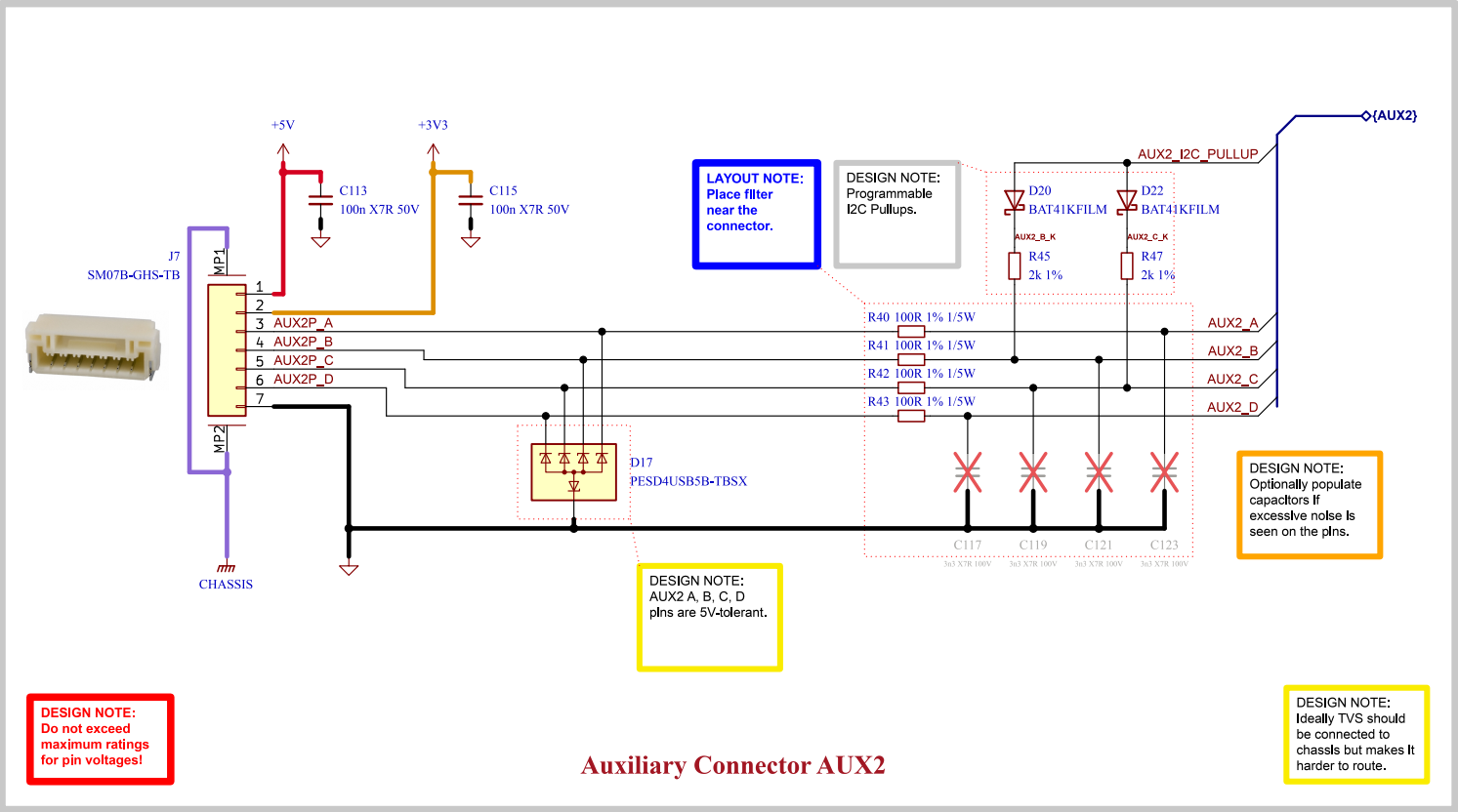
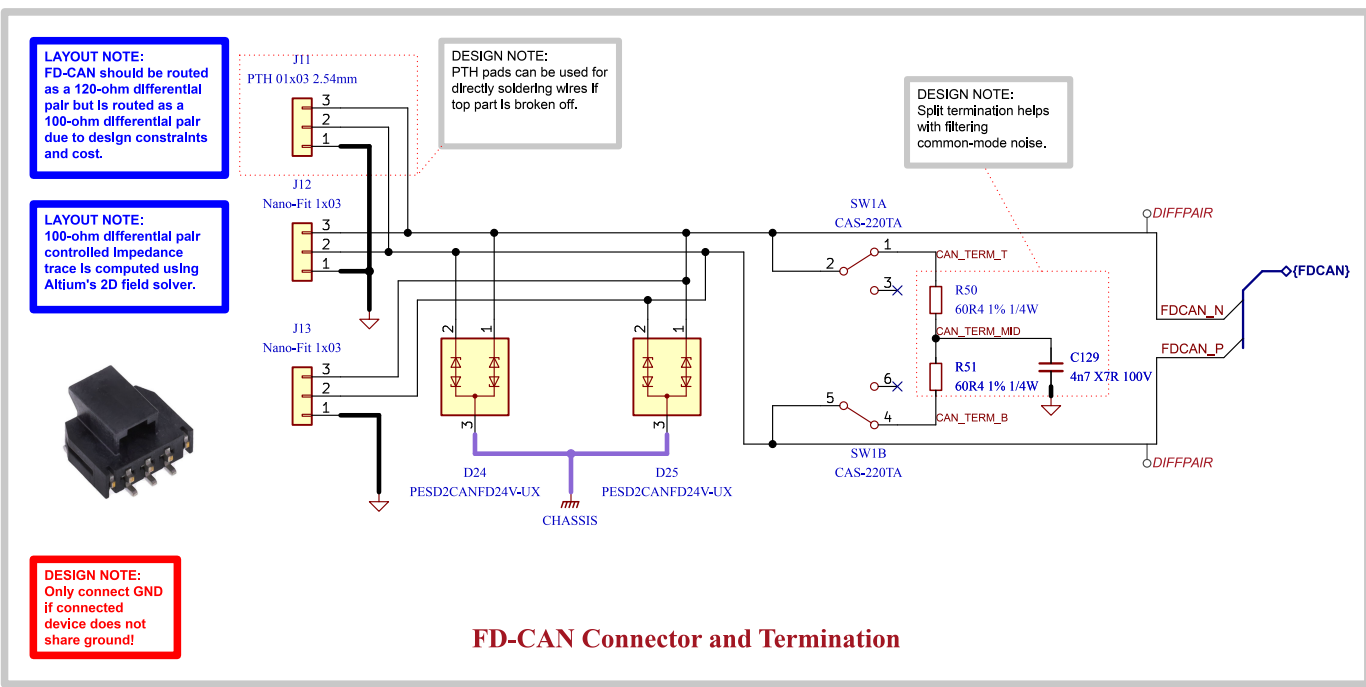
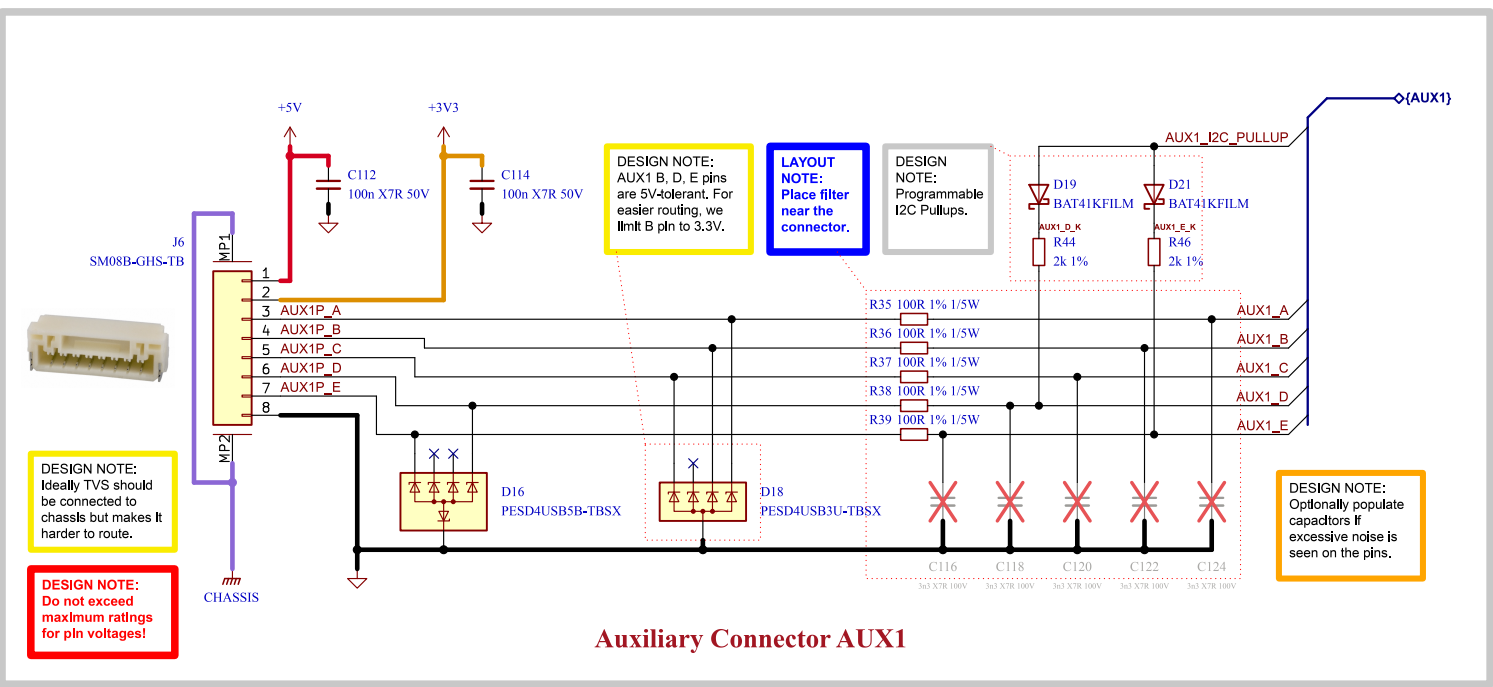
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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - FD-CAN	File Name: Interface - FD-CAN.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1
	Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: A4	Sheet: 16 of 21

[17] Interface - Fan Control



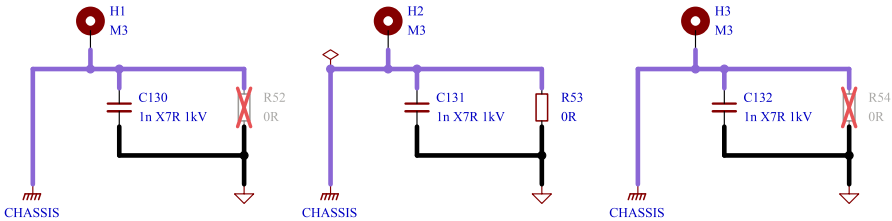
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		Amulet Motion Controller	Chienpanzé		
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Interface - Fan Control	Interface - Fan Control.kicad_sch	Vincent Nguyen	2023-11-19	1.1
	Sheet Path:	Reviewer:	Size:	Sheet:	
	/Project Architecture/Interface - Fan Control/		A4	17 of 21	

[18] Interface - Interconnects



Comments: Reference: Flexible I/O worked examples	Company: EPFL Xplore Research		Variant: RELEASED	Git Hash: 04d3b06
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Interface - Interconnects	File Name: Interface - Interconnects.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31
	Sheet Path: /Project Architecture/Interface - Interconnects/	Reviewer:	Size: A3	Revision: 1.1
			Sheet: 18 of 21	

[19] Misc - Holes, Fiducials



DESIGN NOTE:
Optional RC filter for chassis
grounding. Capacitors offer multi-point
low-impedance connections at
high-frequencies.

DESIGN NOTE:
0-Ohm resistor provides low-impedance
bond between chassis and signal GND.
Be aware of ground loops through cable
shield.

Mounting Holes and Chassis Connection

Front



Back



Fiducial Markers

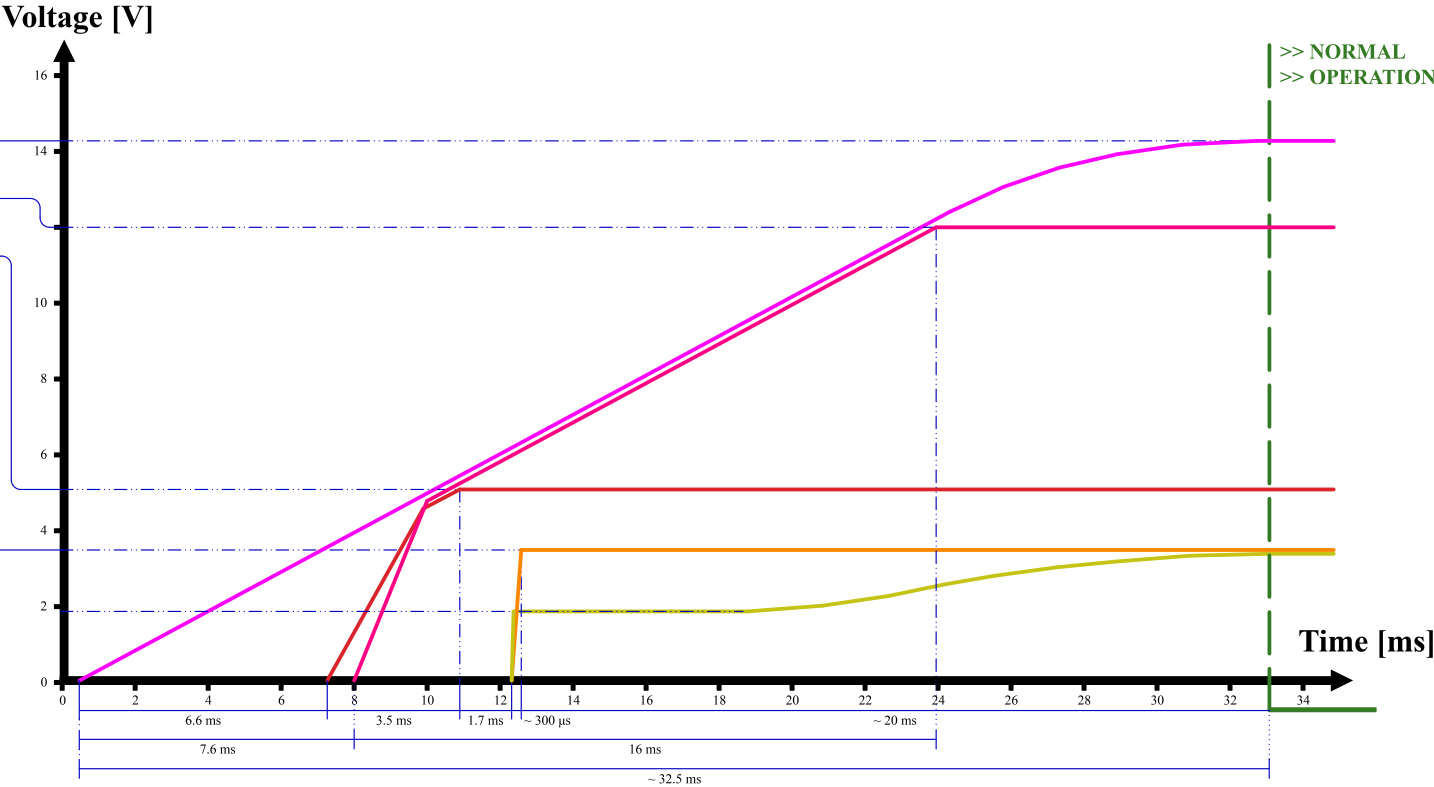
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	Sheet Title: Misc - Holes, Fiducials	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Vincent Nguyen		Date: 2023-10-22	Revision: 1.1	
	Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:		Size: A4	Sheet: 19 of 21	

[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V $\pm$ 1.5%
+5V	LMR36506	5V $\pm$ 1.5%
+3V3	TPS62172	3.3V $\pm$ 3%
+A3V3	LP2992	3.3V $\pm$ 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



	Comments:	Company: EPFL Xplore Research		Variant: RELEASED	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Vincent Nguyen	Date: 2024-03-12	Revision: 1.1
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 20 of 21

[21] Revision History

12-DEC-2023 - Initial Release
Variant: v1.0 Preliminary

- Changed CPH-CPL capacitor to 47nF (gate driver).
- Changed FD-CAN transceiver IC.
- Changed FETs for top cooled variant.
- Added TVS protection and termination switch to FD-CAN.
- Added low-side switched 12V 600mA source for external fan.
- Added LDO for analog supply.
- Changed input power TVS diode to bidirectional and added one diode per connector.
- Moved SOx low-pass filter to MCU section. Should be placed near MCU to avoid noise coupling into ADC lines.
- Added second onboard I2C magnetic encoder for disambiguation.
- Switched PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing.
- Changed RS422 pinout on connector.
- Added ESD protection to all interfaces.
- Added overvoltage protection on thermistor ADC inputs.
- Changed buck regulators to optimize for low noise.
- Added Pi filters to inputs of buck regulators and MCU analog supply.
- Added decoupling caps next to power pins of connectors.

25-JAN-2024 - First Revision
Variant: v1.0 Checked

- Added controller target specifications.
- Replaced 5V 300mA buck converter with 600mA version.
- Added credits to moteus on cover page.
- Added optional RC-Snubber to power stage.
- Increased chassis length to go around the board.
- CAN and power TVS diodes now go to chassis.
- Changed clearance between nets to respect IEC60664-1 where possible.
- Rectified comment on precharge.
- Changed power TVS diode reference designator from "U" to "D".
- Replaced chassis-GND capacitor by 1nF 1kV.

12-MAR-2024 - First Revision
Variant: v1.0 Released

- Modified power sequencing graph according to experimental data.

13-APR-2024 - Second Revision
Variant: v1.1 Released

- Added RC snubber passive values.
- Added more vias for VBUS and LMR36006 GND pads.
- Changed board version voltage reference from +3V3 to +A3V3.
- Changed motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.

	Comments:	Company: EPFL Xplore Research				Variant: RELEASED	Git Hash: 04d3b06
		Board Name: Amulet Motion Controller				Project Name: Chienpanzé	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Vincent Nguyen		Date: 2024-01-03	Revision: 1.1		
Sheet Path: /Revision History/		Reviewer:		Size: A4	Sheet: 21 of 21		